CYBER SECURITY

Internship Project Report

CYBER SECURITY FUNDAMENTALS	NETWORKING & INFORMATION SECURITY

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Certificate / Declaration

I, **Jotishana**, a student of **B.Com 3rd Year**, submit this project report titled **“Cyber Security”** as a learning record for my cyber security internship. The report brings together the fundamental concepts, networking topics, information security principles, ethical hacking concepts, and safe practical activities covered during learning.

This report is prepared for academic purposes. Any practical testing described here should be performed only on systems, networks, and applications that the owner has explicitly authorised.

I would like to thank my instructors and learning resources for introducing me to cyber security and helping me understand how digital systems communicate and how information can be protected. This internship learning has helped me recognise the importance of safe online behaviour, responsible technology use, and continuous skill development.

Abstract

Cyber security is the practice of protecting computers, networks, applications, and information from unauthorised access, misuse, disruption, or damage. This report explains security fundamentals, common threats, ethical hacking, network communication, the OSI and TCP/IP models, IP addressing, subnetting, DNS, internet protocols, Kali Linux, virtualisation, footprinting, network mapping, and web application security. It also includes a case study, safe student activities, a glossary, and a conclusion.

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1. Introduction to Cyber Security

Cyber security means protecting digital devices, systems, networks, and data from cyber threats. It includes technology, processes, and people. Security is not only the responsibility of technical experts: students, employees, organisations, and families all have a role in keeping information safe.

Importance of Cyber Security

- Protects personal information such as identity details, passwords, photos, and financial records.
- Helps businesses maintain customer trust and protect confidential information.
- Supports the availability of services such as online banking, education portals, hospitals, and communication platforms.
- Reduces financial loss, operational disruption, reputational damage, and legal risk.
- Helps protect critical infrastructure and public services.

Cyber Security Fundamentals

The main areas include network security, application security, endpoint security, cloud security, identity and access management, data protection, incident response, and user awareness. A good security programme combines prevention, detection, response, and recovery.

Cyber Security Losses and Impact

A cyber incident may cause stolen money, data loss, downtime, recovery expenses, damaged reputation, privacy violations, and regulatory penalties. The impact depends on what was exposed, how long systems were unavailable, and how quickly the organisation responded.

> *The best approach is layered protection: no single product can stop every threat. Strong passwords, multi-factor authentication, updates, backups, access controls, monitoring, and staff training work together.*

2. Information Security and Its Core Principles

Information security, often called InfoSec, protects information regardless of whether it is stored digitally, printed on paper, or shared verbally. Cyber security focuses more specifically on digital systems and the threats affecting them.

The CIA Triad

- **Confidentiality:** only authorised people should access information. Example: encrypting a confidential student record.
- **Integrity:** information should remain accurate and should not be changed without permission. Example: restricting who can edit accounting records.
- **Availability:** authorised users should be able to access systems when needed. Example: maintaining tested backups and recovery plans.

Other Useful Principles

- **Authentication:** verifying who a user is.
- **Authorisation:** deciding what an authenticated user may do.
- **Accountability:** keeping logs so actions can be traced to responsible accounts.
- **Least privilege:** giving each user only the access needed for their role.
- **Non-repudiation:** using suitable evidence, such as signed records, to help prevent a party from falsely denying an action.

Protecting Personal Data

- Use long, unique passwords and a reputable password manager.
- Turn on multi-factor authentication wherever available.
- Check links and sender addresses before opening messages or attachments.
- Keep operating systems, browsers, and apps updated.
- Use privacy settings carefully and share only necessary personal information.
- Back up important files and avoid unknown USB devices or untrusted downloads.
- Use secure networks; avoid sensitive activity on untrusted public Wi-Fi unless protected by an appropriate secure connection.

Information Security in Organisations

Organisations should classify data, assign access by role, encrypt sensitive records, maintain secure backups, monitor unusual activity, train employees, test incident response plans, and review vendors. Security policies should explain how information is collected, stored, shared, retained, and deleted.

3. Common Threats and How Attacks Happen

A threat is something that could cause harm to a system or its information. A vulnerability is a weakness that could be exploited. Risk is the possibility and impact of a threat exploiting a vulnerability.

Common Types of Threats

- **Phishing:** deceptive messages that try to make people reveal credentials or open harmful links.
- **Malware:** malicious software such as spyware, ransomware, worms, or trojans.
- **Password attacks:** attempts to guess, reuse, or steal passwords.
- **Social engineering:** manipulating people into sharing information or taking unsafe actions.
- **Denial-of-service:** overwhelming a service so legitimate users cannot access it.
- **Insider threat:** risk caused by a staff member or other trusted user, intentionally or accidentally.
- **Unpatched vulnerabilities:** weaknesses that remain because updates or fixes have not been applied.
- **Misconfiguration:** insecure settings, such as publicly exposed storage or excessive permissions.
- **Web application attacks:** attempts to exploit weaknesses in login, input handling, sessions, or access controls.
- **DNS-related attacks:** attempts to misdirect users, abuse DNS infrastructure, or disrupt name resolution.

How a Cyber Attack May Happen: Seven High-Level Stages

- 1. Target selection: an attacker chooses an organisation, account, or service.
- 2. Information gathering: publicly available details may be collected.
- 3. Initial access attempt: for example, a deceptive message or exploitation of a weakness.
- 4. Execution: an unsafe file, script, or vulnerable feature may be triggered.
- 5. Privilege or access expansion: the attacker may try to gain access beyond the original account.
- 6. Impact: data may be exposed, changed, encrypted, or services disrupted.
- 7. Detection and response: defenders investigate, contain the incident, restore systems, and improve controls.

This is a simplified defensive overview, not a guaranteed sequence. Real incidents may skip stages or occur in a different order. Organisations use monitoring, patching, access control, backups, and staff awareness to interrupt attacks early.

Warning Signs / Red Flags

- Unexpected password reset messages or multi-factor prompts.
- Unusual sign-ins, unknown devices, or login alerts.
- Unexpected changes to files, account settings, or payment details.
- Devices running unusually slowly or displaying unfamiliar security warnings.
- Requests for urgent payments, secrecy, gift cards, or confidential data.

4. Ethical Hacking, Careers, Skills and Tools

Introduction to Ethical Hacking

Ethical hacking is authorised security testing performed to find weaknesses before criminals exploit them. Ethical hackers agree on the scope, obtain written permission, protect collected data, avoid unnecessary disruption, and report findings responsibly.

Good Hackers versus Bad Hackers

Ethical hacker	Malicious hacker
Works with permission and a defined scope.	Accesses systems without permission.
Documents weaknesses and recommends fixes.	May steal, damage, extort, or misuse information.
Protects evidence and respects privacy.	May hide activity and disregard privacy.

Types of Ethical Hacking / Security Testing

- Network testing: reviews network devices, segmentation, and exposed services.
- Web application testing: checks authentication, authorisation, input validation, and session handling.
- Mobile application testing: reviews app permissions, data storage, and communication security.
- Wireless security testing: checks Wi-Fi configuration with explicit permission.
- Cloud security review: checks identity, permissions, logging, and storage configuration.
- Social-engineering assessment: evaluates awareness only under an approved, carefully controlled plan.

Skills Required to Become an Ethical Hacker

- Networking: IP addresses, ports, DNS, routing, TCP/IP, and firewalls.
- Operating systems: Linux and Windows basics, permissions, processes, and logs.
- Scripting: beginner Python, Bash, or PowerShell to automate safe tasks.
- Web fundamentals: HTTP/HTTPS, cookies, sessions, APIs, and databases.
- Security concepts: vulnerabilities, risk, encryption, authentication, and incident response.
- Analytical thinking, careful documentation, communication, patience, and ethical judgement.

Tools and Platforms Used in Ethical Hacking

- **Wireshark:** examines network traffic in a lab or on an authorised network.
- **Nmap:** discovers hosts and services within an approved scope.
- **Burp Suite Community Edition:** helps inspect web requests in a legal test environment.
- **OWASP Juice Shop / WebGoat:** intentionally vulnerable training applications for safe practice.
- **Kali Linux:** a Linux distribution that includes many security assessment tools.

- **VirtualBox or VMware:** runs isolated virtual machines for labs.
- **Git and GitHub:** records project files, notes, scripts, and documentation.

Tools do not make someone an ethical hacker by themselves. Permission, safe scope, sound methodology, and responsible reporting are essential.

5. Network Basics: How Computers Communicate

A computer network is a group of connected devices that exchange information. Devices use agreed rules called protocols so that data can be addressed, transmitted, received, and interpreted.

Basic Networking Terms

- **Node:** any device connected to a network.
- **Router:** forwards traffic between different networks.
- **Switch:** connects devices within a local area network.
- **Access point:** provides wireless network connectivity.
- **Modem:** connects a home or office network to an internet service, depending on the technology.
- **Firewall:** allows or blocks traffic according to security rules.
- **Packet:** a formatted unit of data carried across a network.
- **Port:** a number used to identify a service or application endpoint.

Internal Network Basics

A local area network (LAN) connects devices in a limited area, such as a home, office, or campus. A wide area network (WAN) connects networks across larger distances. Wired Ethernet and Wi-Fi are common ways to connect devices.

The Language of Networks

Protocols define how devices communicate. Examples include IP for addressing and routing, TCP for reliable ordered delivery, UDP for lightweight datagram delivery, DNS for translating domain names, HTTP for web communication, and TLS for protecting data in transit.

6. OSI Model and TCP/IP Model

The Seven Layers of the OSI Model

Layer	Name	Main purpose / example
7	Application	Services used by applications; HTTP, DNS
6	Presentation	Data format, encoding, encryption concepts
5	Session	Managing communication sessions
4	Transport	End-to-end delivery; TCP, UDP
3	Network	Logical addressing and routing; IP
2	Data Link	Local frames and MAC addresses; Ethernet, Wi-Fi link
1	Physical	Signals, cables, radio, connectors

What Is TCP/IP?

TCP/IP is the protocol suite used by the internet and many private networks. IP helps address and route packets; TCP provides reliable, ordered delivery where needed. TCP/IP is practical for real network communication, while the OSI model is often used as a learning and troubleshooting framework.

The Four Layers of TCP/IP

- **Application:** application protocols such as HTTP, DNS, SMTP, and SSH.
- **Transport:** TCP or UDP communication between application endpoints.
- **Internet:** IP addressing and routing between networks.
- **Network access / link:** local delivery over Ethernet, Wi-Fi, and related technologies.

TCP/IP versus OSI

The OSI model has seven conceptual layers. The common four-layer TCP/IP model groups some OSI functions together. The models are not competitors: OSI helps explain concepts, while TCP/IP describes the protocol suite used in real networks.

Example: Sending a WhatsApp Message

A messaging app prepares the content, uses network protocols to send encrypted traffic to the service, and the receiving device obtains and processes it. The exact implementation may change, but the general journey includes application logic, transport, IP routing, and local network transmission. End-to-end encryption protects message content in supported chats, while metadata and backups may have separate protections and settings.

7. IP Addresses, NAT and Subnetting

What Is an IP Address?

An IP address is a logical address used to identify an interface on an IP network. IPv4 uses 32 bits, commonly written as four decimal numbers such as 192.0.2.10. IPv6 uses 128 bits and is written in hexadecimal groups.

Public versus Private IP Addresses

A public IP address is globally routable on the internet. Private IPv4 ranges are reserved for internal networks and are not routed across the public internet in the usual way.

Private IPv4 range	Typical size
10.0.0.0 – 10.255.255.255	10.0.0.0/8
172.16.0.0 – 172.31.255.255	172.16.0.0/12
192.168.0.0 – 192.168.255.255	192.168.0.0/16

NAT: Network Address Translation

NAT allows a router or gateway to translate addresses, often letting multiple private IPv4 devices share one public IPv4 address. NAT is not a replacement for a firewall or other security controls.

Static versus Dynamic IP Addresses

A static address is intended to remain the same; a dynamic address is assigned and may change, often through DHCP. Either approach can be secure when configured and managed correctly.

What Is Subnetting?

Subnetting divides an IP network into smaller logical networks. It helps organise devices, use address space efficiently, separate departments or systems, and support routing and security controls.

Subnet Masks and CIDR

A subnet mask identifies the network portion and host portion of an IPv4 address. CIDR notation uses a slash followed by the number of network bits, such as /24. A /24 IPv4 network has 256 total addresses; in a conventional IPv4 subnet, 254 are usable host addresses because the network and broadcast addresses are reserved.

Simple Example

For 192.168.1.0/24, the network address is 192.168.1.0, the conventional broadcast address is 192.168.1.255, and the usual host range is 192.168.1.1 to 192.168.1.254. Real designs should account for the exact subnet size, gateways, reserved addresses, and IPv6 where applicable.

Why Subnetting Matters for Security

Separating guest Wi-Fi, staff devices, servers, and management systems into appropriate network segments can limit unnecessary communication. Segmentation should be combined with firewall rules, identity controls, monitoring, and secure configuration.

8. DNS: The Internet Phone Book

What Is DNS?

The Domain Name System (DNS) translates human-readable domain names, such as example.com, into records that computers use to locate services. It is often described as the internet phone book, although it stores several kinds of records, not only addresses.

DNS Hierarchy and Main Players

- **Root servers:** direct resolvers toward the relevant top-level domain.
- **TLD servers:** handle domains such as .com, .org, or country-code domains.
- **Authoritative name servers:** publish the official DNS records for a domain.
- **Recursive resolver:** looks up answers for a client and may cache results.
- **Stub resolver:** the client-side component that sends queries to a configured resolver.

How DNS Resolution Works

- 1. A device checks relevant local caches and configuration.
- 2. If needed, it asks a recursive resolver.
- 3. The resolver follows referrals through the DNS hierarchy or uses cached data.
- 4. The authoritative server provides the requested record.
- 5. The resolver returns the result and caches it for the record's remaining TTL.

Common DNS Record Types

- **A:** maps a name to an IPv4 address.
- **AAAA:** maps a name to an IPv6 address.
- **CNAME:** aliases one name to another name.
- **MX:** identifies mail servers for a domain.
- **NS:** identifies authoritative name servers.
- **TXT:** stores text used for verification and email-security policies, among other purposes.
- **PTR:** supports reverse lookups from addresses to names.
- **SOA:** contains key information about a DNS zone.

Caching and TTL

A Time to Live (TTL) tells resolvers how long a record may be cached. Caching improves speed and reduces repeated queries, but changes may not be visible everywhere until cached data expires.

DNS Security and Common Attacks

Risks include spoofed or poisoned responses, DNS hijacking, misuse of DNS for phishing or malware communication, and denial-of-service attacks. Defences include secure registrar accounts, multi-factor authentication, careful access control, DNSSEC validation where supported, monitoring, resilient providers, and accurate record management.

Safe DNS Tools

For domains you own or are authorised to examine, tools such as nslookup, dig, and online DNS lookup services can help inspect records. Avoid changing records or scanning systems without permission.

9. Internet Protocols and the Journey to a Website

What Happens When You Type a Website Address?

- 1. The browser interprets the URL and checks relevant cached information.
- 2. DNS resolution finds the address for the domain when required.
- 3. The device establishes the appropriate network connection; HTTPS commonly uses TLS to protect traffic.
- 4. The browser sends an HTTP request.
- 5. The server processes the request and returns a response.
- 6. The browser renders the page and may request additional images, scripts, and stylesheets.

HTTP versus HTTPS

HTTP is the protocol used to request and deliver web resources. HTTPS is HTTP protected by TLS, which helps provide confidentiality, integrity, and server authentication when correctly configured. A padlock does not prove that a website is honest; users must still check the domain and be alert to scams.

Other Important Internet Tools and Protocols

- **DHCP:** automatically provides network configuration such as IP addresses.
- **ARP / Neighbor Discovery:** helps map IP-layer destinations to local-link addresses, depending on IP version.
- **ICMP:** carries network control and diagnostic messages; ping commonly uses ICMP.
- **SSH:** supports secure remote administration.
- **SMTP, IMAP, POP3:** protocols used in email delivery and access.
- **TLS:** protects many types of network communication.
- **VPN:** creates a protected tunnel across an underlying network, depending on its configuration.

Real-World Applications and Practical Labs

Networks support online banking, video calls, e-commerce, learning platforms, cloud storage, smart devices, and business applications. Safe practical exercises include viewing your own device's IP configuration, looking up DNS records for a public domain, and observing sample traffic in a local lab. Do not capture other people's private traffic or test networks without permission.

10. Kali Linux, Virtualisation, Footprinting and Network Mapping

What Is Kali Linux?

Kali Linux is a Linux distribution designed for penetration testing, digital forensics, and security research. It includes many security tools, but it should be used only within an authorised lab or approved engagement.

What Is Virtualisation?

Virtualisation allows one physical computer to run isolated virtual machines. A student can create a lab with a security-testing machine and intentionally vulnerable practice target without exposing real systems. Keep lab networks isolated, use snapshots, and avoid bridged networking unless the purpose and risks are understood.

What Is Footprinting?

Footprinting is the process of gathering information about a target environment. Passive footprinting uses public information without directly interacting with the target's systems; active footprinting interacts with systems and requires explicit permission.

Why Footprinting Matters

Defenders use authorised information gathering to understand their public exposure, identify outdated assets, review domain and certificate information, and reduce unnecessary disclosure. Only gather information that is relevant to a legitimate purpose.

Network Mapping

Network mapping creates an overview of authorised devices, subnets, services, and connections. Organisations use maps to maintain inventories, understand traffic paths, plan segmentation, and investigate incidents. Maps should be access-controlled because they can reveal sensitive infrastructure details.

Safe Practical Lab Plan

- Create an isolated virtual machine and take a snapshot before testing.
- Use a deliberately vulnerable training application such as OWASP Juice Shop in the lab.
- Write down the lab scope, learning goal, and safety limits.
- Observe basic network settings and DNS queries generated by your own test device.
- Record the finding, evidence, risk, and a defensive recommendation.
- Restore the snapshot and remove temporary test data after the exercise.

11. Web Application Security Fundamentals

Web applications combine a browser or client, a server, APIs, databases, identity systems, and third-party services. A weakness in any component may affect the confidentiality, integrity, or availability of data.

Common Web Application Weaknesses

- **Broken access control:** users can view or change information outside their permissions.
- **Injection:** untrusted input is interpreted as part of a command or query.
- **Authentication weaknesses:** login and account recovery controls are inadequate.
- **Security misconfiguration:** unnecessary services, default settings, or overly broad permissions remain enabled.
- **Cryptographic failures:** sensitive information is not adequately protected.
- **Cross-site scripting (XSS):** untrusted content is executed in a user's browser.
- **Vulnerable components:** dependencies contain known weaknesses or are no longer supported.
- **Insufficient logging and monitoring:** suspicious activity is missed or investigated too late.

Defensive Practices

- Validate input and use safe parameterised database queries.
- Enforce server-side authorisation on every sensitive action.
- Use secure session handling, multi-factor authentication, and safe recovery procedures.

- Encrypt data in transit and protect sensitive data at rest.
- Patch dependencies and remove unused components.
- Apply secure headers, rate limits, logging, monitoring, and tested backups.
- Use code review and authorised security testing before release.

A vulnerability report should describe the affected component, the risk, evidence collected safely, likely impact, and a clear remediation step. Avoid exposing real personal data in screenshots or reports.

12. Cyber Security Certifications and Learning Programmes

Careers in Cyber Security

- **Security analyst:** monitors alerts, investigates suspicious activity, and supports incident response.
- **Network security administrator:** configures and maintains firewalls, segmentation, and network controls.
- **Penetration tester:** conducts authorised testing and reports exploitable weaknesses.
- **Security engineer:** designs and implements protective systems.
- **Digital forensics analyst:** examines digital evidence using documented procedures.
- **Cloud security specialist:** protects cloud identities, workloads, storage, and configurations.
- **Governance, Risk and Compliance (GRC) analyst:** supports policies, audits, risk assessments, and compliance.

Global Cyber Security Certificates

Certification names, eligibility, fees, and exam formats can change, so always verify current details on the issuing organisation's official website. Examples of widely recognised learning and certification pathways include:

- **ISC2 Certified in Cybersecurity (CC):** an entry-level security credential; check current eligibility and offer details.
- **CompTIA Security+:** covers broad foundational security concepts and operational practices.
- **CompTIA Network+:** focuses on networking fundamentals useful for security roles.
- **Cisco CCST Cybersecurity / Networking:** entry-level learning pathways; verify the current exam names and requirements.
- **Certified Ethical Hacker (CEH):** focuses on ethical hacking concepts and security testing; check the provider's current requirements.
- **GIAC certifications:** specialised credentials across areas such as security operations, incident handling, and penetration testing; often more advanced and costly.

A certificate can demonstrate structured learning, but it does not replace hands-on practice, ethical conduct, communication, or experience. Choose a pathway based on your current skills, budget, and target role.

Future Skills Edge Programme

The name “Future Skills Edge” may refer to a particular training programme or platform. Its exact syllabus, certification, and recognition should be confirmed from the provider or internship material. In general, future-skills programmes aim to build digital literacy, problem-solving, communication, collaboration, adaptability, and job-related technology skills. Add the programme's official provider, modules, and completion evidence here if your internship supplied them.

Emerging Trends and Technologies

- Cloud security and identity-first security.
- Artificial intelligence for detection, triage, and security automation, alongside new AI-related risks.
- Zero Trust: continuously evaluating identity, device, and access context rather than assuming a network is trusted.
- Secure software development and software supply-chain protection.

- Privacy engineering, data governance, and stronger authentication.
- Internet of Things (IoT) security and protection of connected devices.
- Post-quantum cryptography planning as standards and migration needs evolve.

13. Case Study and Student Activities

Case Study: A Phishing Email at a College

Situation: A student receives an email claiming that their college account will be disabled unless they sign in through an unfamiliar link. The message uses urgent language and has an unusual sender address.

Risks

- The student may disclose their password on a fake sign-in page.
- An attacker may access email, personal files, or linked services.
- The account could be used to send further deceptive messages.

Recommended Response

- Do not click the link or enter credentials.
- Report the message through the college's official IT or phishing-reporting channel.
- If credentials were entered, change the password using the official site and notify IT promptly.
- Enable multi-factor authentication and review recent account activity.
- The organisation should block malicious domains where appropriate, investigate reports, and provide awareness training.

Quick Student Activity

Activity 1 — Spot the red flags: review a fictional email and list three warning signs. Activity 2 — CIA triad: identify whether a scenario mainly affects confidentiality, integrity, or availability. Activity 3 — Network journey: draw the steps from entering a website name to receiving a page. Activity 4 — Personal security checklist: review password uniqueness, MFA, updates, backups, and privacy settings. Use fictional examples only.

Mini Practical: DNS Lookup

Goal: understand how DNS records are returned for a domain. In a permitted learning environment, use a normal DNS lookup utility for a public domain, record the query type and result, and explain what an A, AAAA, or MX record means. Do not attempt to alter records or disrupt DNS services.

Sample Risk Table

Risk	Possible impact	Basic control
Phishing	Account compromise	Awareness, MFA, reporting
Outdated software	Known weakness exploited	Patch management
Lost device	Data exposure	Screen lock, encryption, remote management
Data deletion	Service or work disruption	Backups and restore tests

14. Conclusion, Key Takeaways and Glossary

Cyber security is a shared responsibility that requires technology, good processes, and informed people. This project introduced the main principles of information security, common threats, ethical hacking, networking, IP addressing, subnetting, DNS, internet protocols, virtualisation, and web application security. The most important lesson is to practise responsibly: use authorised environments, protect data, document findings clearly, and focus on preventing harm.

Information Security Closing Takeaways

- Protect confidentiality, integrity, and availability.
- Use unique passwords, multi-factor authentication, updates, and backups.
- Understand networking fundamentals before moving to advanced security testing.
- Use tools only with permission and within an agreed scope.
- Report security issues responsibly and recommend practical fixes.
- Keep learning: threats, technologies, and defensive practices change over time.

Glossary

Asset	Anything valuable that needs protection.
Attack surface	The places where a system could be exposed to threats.
Authentication	Checking a user's identity.
Firewall	A control that filters network traffic using rules.
IP address	A logical address used for communication on IP networks.
Malware	Software designed to cause harm or perform unauthorised actions.
MFA	Multi-factor authentication; using more than one type of proof.
Packet	A unit of formatted data sent across a network.
Patch	An update that fixes bugs or security weaknesses.
Phishing	A deceptive attempt to obtain information or trigger unsafe action.
Subnet	A logical subdivision of an IP network.
Vulnerability	A weakness that could be exploited.
Zero Trust	A security approach that continually verifies access and context.

15. References and Learning Resources

The following reputable resources can support further study. Check each website for the latest documentation, course details, and certification requirements.

- National Institute of Standards and Technology (NIST) — Cybersecurity Framework and security publications: https://www.nist.gov/cyberframework
- Cybersecurity and Infrastructure Security Agency (CISA) — security guidance and awareness: https://www.cisa.gov/
- OWASP — web application security guidance and training projects: https://owasp.org/
- IETF RFC Editor — internet standards and protocol specifications: https://www.rfc-editor.org/
- ICANN — domain names and DNS information: https://www.icann.org/
- Kali Linux official documentation: https://www.kali.org/docs/

- Wireshark documentation: https://www.wireshark.org/docs/
- CompTIA certifications: https://www.comptia.org/certifications
- ISC2 certifications: https://www.isc2.org/certifications
- Cisco learning and certifications: https://www.cisco.com/
- EC-Council certification information: https://www.eccouncil.org/

Note: This report is an educational overview, not a claim that a formal certification was awarded or that any penetration test was conducted. Replace or supplement programme-specific details with verified information from your internship provider.